

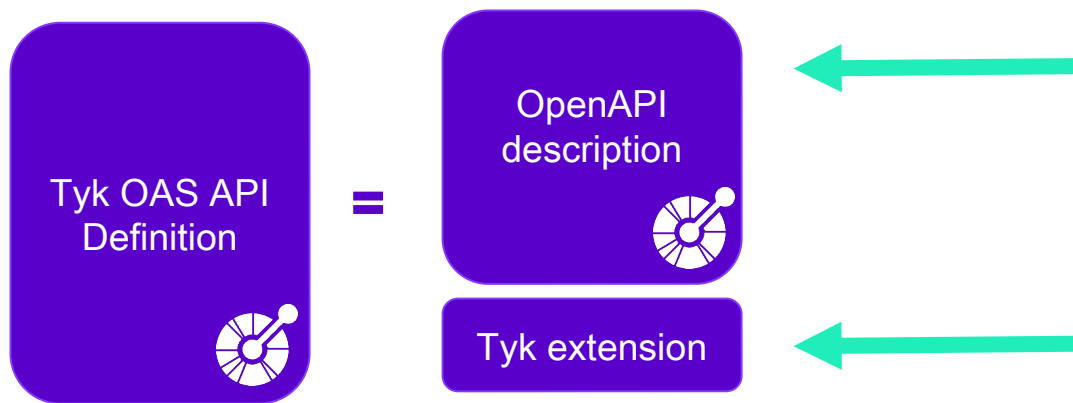
Tyk OAS API Definitions

Tyk OAS Definition

- Standard of defining APIs via OpenAPI Specification
- Tyk supports the importing and configuration on the OAS
- Significant time and complexity savings if you are already using it to design and document your APIs



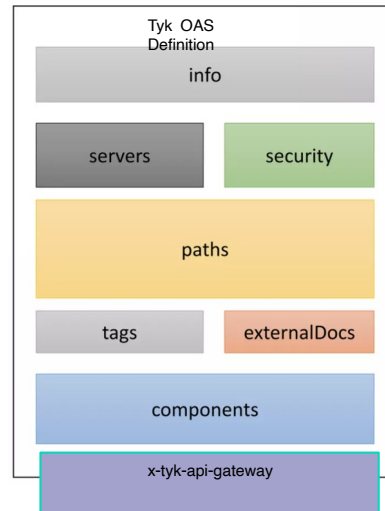
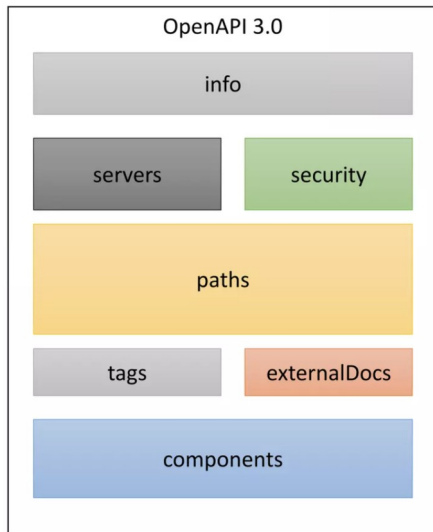
What's Tyk OAS API definition?



From your API design tool
or code editor
(info, servers, paths,
components, ...)

Export from Dashboard or
Start from a template
(gateway configurations,
middleware configurations,
listenPath, metadata, ...)

Tyk OAS Definition



Tyk Vendor Extension

Reference

Custom Tyk API extensions inside the OpenAPI definition.

Stored under the key `x-tyk-api-gateway`.

Main Fields

- `info` (Info)
 - Main metadata for the API definition
- `upstream` (Upstream)
 - Configurations related to the upstream target
- `server` (Server)
 - Configurations for the API server
- `middleware` (Middleware)
 - Configurations for Tyk middleware

Use `x-tyk-api-gateway` to enrich your OpenAPI spec with Tyk-specific runtime and security controls.

Tyk OAS Definition

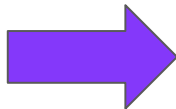
- Authentication

- Auth token
- JWT
- OAuth 2.0
- mTLS
- Basic Auth
- Multi-Auth

☐ Apply Detected Authentication

OAS definitions contain information about how the API should be protected. Tyk will read and apply the relevant security policies.

```
"components": {
  "securitySchemes": {
    "authToken": {
      "type": "apiKey",
      "name": "Authorization",
      "in": "header"
    }
  }
},
"security": [
  {
    "authToken": []
  }
],
"paths": {
  "/example": {
    "get": {
      "summary": "Example endpoint",
      "security": [
        {
          "authToken": []
        }
      ]
    }
  }
],
```



```
"authentication": {
  "enabled": true,
  "securitySchemes": {
    "authToken": {
      "enabled": true
    }
  }
}
```

```
"paths": {
  "/example": {
    "get": {
      "operationId": "exampleGET",
      "responses": {
        "200": {
          "description": "Successful response"
        }
      },
      "security": [
        {
          "authToken": []
        }
      ],
      "summary": "Example endpoint"
    }
  }
},
```

Tyk OAS Definition

- Validation

- Automatically creates a request validation middleware in Tyk with schema in OAS

☐ Generate Validate Request Middleware

For all paths defined in the OAS definition that have a requestBody schema defined, Tyk will read and validate any request against the JSON schema provided.

JSON Validation

The request body will be validated by using this OAS schema

OAS Schema

```
1 {  
2   "properties": {  
3     "name": {  
4       "type": "string"  
5     },  
6     "tag": {  
7       "type": "string"  
8     },  
9   },  
10  "required": [  
11    "name"  
12  ],  
13  "type": "object"  
14 }
```

```
"post": {  
  "description": "Creates a new pet in the store. Duplicates are allowed",  
  "operationId": "addPet",  
  "requestBody": {  
    "description": "Pet to add to the store",  
    "required": true,  
    "content": {  
      "application/json": {  
        "schema": {  
          "$ref": "#/components/schemas/NewPet"  
        }  
      }  
    }  
  }  
}
```

Tyk OAS Definition

- Middleware
 - Uses operationIDs for middleware configured for each path
 - Middleware declared in x-tyk-api-gateway section

```
"middleware": {  
  "operations": {  
    "addPet": {  
      "validateRequest": {  
        "enabled": true,  
        "errorCode": 422  
      }  
    }  
  }  
}
```

```
"post": {  
  "description": "Creates a new pet in the store. Duplicates are allowed",  
  "operationId": "addPet",  
  "requestBody": {  
    "content": {  
      "application/json": {  
        "schema": {  
          "$ref": "#/components/schemas/NewPet"  
        }  
      }  
    }  
  },  
  "description": "Pet to add to the store",  
  "required": true  
},
```